



Optimizing camera trap placement for felid density estimation: Comparing trail-based and random deployments

Lungten Dorji^{a,b,*}, C. Patrick Doncaster^a, Wai-Ming Wong^c, Rinjan Shrestha^d, Kuenley Tenzin^e, Paul Glover-Kapfer^f, Lon Grassman Jr.^j, Oliver R. Wearn^{g,h,i}

^a School of Biological Sciences, University of Southampton, United Kingdom

^b Nature Conservation Division, Department of Forests and Park Services, Royal Government of Bhutan, Bhutan

^c Panthera, New York, United States of America

^d WWF-Canada, Toronto, ON, M5V 1S8, Canada

^e World Wildlife Fund, Bhutan

^f Independent

^g Institute of Zoology, Zoological Society of London, London, UK

^h Fauna & Flora, Vietnam Programme, Hanoi, Viet Nam

ⁱ Snow Leopard Trust, Seattle, USA

^j ICF, 5 Lakeway Centre Court, Suite 200, Austin, TX, 78734, USA

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ABSTRACT

Wild felids are globally threatened, requiring accurate monitoring for adaptive conservation. Camera-trap surveys for individually identifiable felids place cameras on trails and estimate density using spatially explicit capture–recapture (SCR) models. However, unequal individual capture probabilities—individual heterogeneity—can bias density estimates if unaccounted for. We examined how camera placement influences SCR-derived density estimates across 400 km² study area in Central Bhutan inhabited by five identifiable felid species (*Panthera tigris*, *Panthera pardus*, *Neofelis nebulosa*, *Pardofelis marmorata*, and *Prionailurus bengalensis*). We deployed 100 paired cameras on trails and 100 single cameras at random locations.

Trail cameras produced a male-biased sex ratio of 1.75 males per female, whereas random cameras yielded a balanced ratio of 1.0. The trail placement missed proportionally more females (30%) than males (8%), while the random placement missed more males (56%) than females (33%). This male-biased detection pattern was consistent across species, particularly for leopard and leopard cat. Density estimates based solely on trail data were, on average, 22% lower than estimates derived from the combined trail + random dataset.

We evaluated two approaches to address this sex bias. First, incorporating sex as a covariate in SCR models produced trail-only estimates comparable to those from the combined dataset. Second, we tested a mixed deployment approach (trail and random) using data sub-sampling, which indicated that allocating ~50% of cameras to random locations provided an effective balance between accuracy and precision. Where sex-specific trail use affects detectability, we recommend mixed designs combining trail and random camera placements, although smaller surveys may still justify trail-only designs.

1. Introduction

The effective conservation and monitoring of wild felid populations, many of which are globally threatened with extinction, requires accurate information about their population sizes (Shi et al., 2023; Macdonald et al., 2010). This is critical for assessing population status,

prioritising sites for conservation, and determining the effectiveness of interventions. Camera traps have emerged as an important tool for population assessments of a wide range of terrestrial mammals and are increasingly being adopted for the monitoring of other taxa (O'Brien and Kinnaird, 2008). Owing to their rarity and cryptic behaviour, felids are notoriously difficult to monitor, and although camera trapping has

* Corresponding author at: Nature Conservation Division, Department of Forests and Park Services, Ministry of Energy and Natural Resources, Royal Government of Bhutan, Thimphu, 11001, Bhutan.

E-mail address: Lungten.Dorji@soton.ac.uk (L. Dorji).

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existed for more than a century (Wearn and Glover-Kapfer, 2017), its use in monitoring skyrocketed following adoption of camera trap photographs to construct individual detection histories to fit capture-recapture models for population size (Karanth and Nichols, 1998). The development of this statistical framework, coupled with that of digital cameras, has made camera trapping the preeminent method for monitoring endangered populations of felids (Harmsen et al., 2020; Wearn and Glover-Kapfer, 2017; Foster and Harmsen, 2012; Karanth and Nichols, 1998; Karanth et al., 2006). Indeed, in combination with the establishment of capture-recapture modelling and, later, spatially explicitly capture-recapture (SCR) modelling, camera trap surveys are now the “gold-standard” method for assessing populations of all felid species that are individually identifiable (Tourani, 2022; Green et al., 2020). Such surveys can provide density estimates, demographic parameters (using ‘open’ population models) and estimation of movement, even for very elusive and endangered wild felids (Palmero et al., 2023; Tourani, 2022). SCR has now been applied to over 150 species, most notably wild felids such as leopards *P. pardus* and tigers *P. tigris* (Tourani, 2022). It is also the standard method used in national surveys of tigers, leopards, and snow leopards *P. uncia* (Penjor et al., 2019; Tempa et al., 2019; Green et al., 2020).

Classical capture-recapture (CR) models often begin with the simplifying assumption of equal capture probability across all individuals within the sampled area (Pollock, 1982; Otis et al., 1978). However, it is well recognized that this “homogeneity” assumption may not hold in real populations, and many CR estimators have been developed to relax it. Heterogeneity in capture probability can be accommodated via methods including finite-mixture models (Otis et al., 1978; Pledger and Efford, 1998; Royle et al., 2009; Gimenez et al., 2018; Stevenson et al., 2022). Spatial capture-recapture (SCR) further incorporates a source of heterogeneity by explicitly modelling how detection probability declines with distance from an individual's activity center (i.e., trap exposure heterogeneity; Borchers and Efford, 2008; Efford, 2014). That said, additional unmodeled heterogeneity remains a concern: if individuals differ in baseline detectability (for reasons unrelated to spatial proximity to traps) or movement behaviour (e.g., home range size), then density estimates can still be negatively biased (due to positive bias in parameters related to capture probability) when that variation is not captured in the model (Efford and Fewster, 2013; Royle, 2011; Gimenez et al., 2018). In felid camera trap surveys, sex differences in space use or trail affinity are classic drivers of such variation (Sollmann et al., 2011; Kolowski and Forrester, 2017; Greco et al., 2025), and thus merit explicit modelling or survey designs that mitigate their effects.

Heterogeneity in individual capture probabilities appears to be particularly high in camera trap studies of big cats (Piña-Covarrubias et al., 2023; Singh et al., 2014; Sollmann et al., 2011). This is often due to differences in capture probabilities between males and females, in turn due to differences in the biology and behaviour of the two sexes and their differing life history patterns. Specifically, given the polygynous social structure of felids in which male home ranges are larger than, and overlap several, female home ranges, we might expect males to preferentially use trails or other easy travel routes (e.g. riparian corridors) to more efficiently patrol their territories and patrol as many female home ranges as possible. In felids, female home ranges and movements are generally smaller than those of males (Morato et al., 2016; McNitt et al., 2020) and female avoidance of areas frequented by males might be expected to prevent harassment and infanticide of young by males (Stasiukynas et al., 2022; Harmsen et al., 2017; Wearn et al., 2013). Perhaps because of these factors, male-biased detection is often observed in camera traps deployed with the typical, trail-based deployment of camera traps (Balme et al., 2013; Harmsen et al., 2017; Srbeek-Araujo, 2018; Singh et al., 2014). Meanwhile, studies that have placed cameras ‘off-trail’ have reported females that are missed by the trail cameras, for example in Sunda clouded leopards *Neofelis diardi* (Wearn et al., 2013) and jaguars (Montalvo et al., 2023). If this

heterogeneity in individual detection probabilities is not accounted for, we would tend to expect negative bias in the estimates of density, as well as occupancy, resulting from such camera trap surveys (Royle and Dorazio, 2006, 2008; Harmsen et al., 2020).

In practice, when sex or other biologically relevant covariates cannot be determined for all photographed individuals, a hybrid mixture model (implemented through the *hcov* argument in the R package *secr*; Efford, 2023) provides a robust alternative to standard finite mixtures. This approach allows known group memberships (e.g., sex identified from genitalia in camera trap images) to be specified explicitly, while estimating group memberships probabilistically for individuals of unknown sex. Although the resulting mixing parameter should not be interpreted as a true sex ratio (Mohamed et al., 2021), hybrid mixture models effectively remove bias in density and abundance estimates caused by unmodelled heterogeneity (Sollmann et al., 2011; Efford, 2014). Potential multimodality in the likelihood can be addressed by employing alternative optimisation algorithms—such as Nelder–Mead or simulated annealing—rather than the default Newton–Raphson routine (Efford, 2014). In this study, we therefore used the hybrid mixture framework to accommodate uncertain sex assignments, while also exploring how complementary adjustments to survey design (i.e., combining trail-based and random camera placement) can further minimise heterogeneity and improve robustness of density estimates.

An alternative is to attempt to reduce heterogeneity through random camera placement. In this approach, camera traps would be set randomly with respect to habitat, and therefore heterogeneity caused by habitat, and its interaction with individual characteristics such as sex, would be much reduced. Several studies have shown that incorporating random or spatially representative camera placements—rather than relying solely on trail-based designs—can improve the accuracy and reduce bias of population estimates by sampling individuals with differing movement behaviours or habitat preferences (Kolowski and Forrester, 2017; Tanwar et al., 2021; Fonteyn et al., 2021; Tourani, 2022). Such an approach would be expected to better expose all individuals in a population to sampling and, insofar as capture heterogeneity is caused by differences in habitat-use among individuals, better equalise detections across individuals. Density estimates resulting from a random survey should, therefore, be more accurate than from surveys using trail-focused (or otherwise habitat-restricted) camera surveys. Unlike analytical frameworks such as distance sampling, CR/SCR methods do not require strict spatial randomization because the primary objective is to sample individuals rather than space. Nevertheless, reducing systematic heterogeneity in exposure to sampling remains desirable because strong behavioural or habitat-related differences in detectability can bias parameter estimates if left unmodelled. The main drawback of a randomised survey design, and the reason it has not been used more widely, is that there is a risk of capture rates being substantially lower, which may increase parameter uncertainty or, in the worst case, preclude parameter estimation altogether. The main strategy to avoid this loss of precision is to sample with greater intensity (number of locations and duration), but resources may in many cases be limited. A hybrid approach of combining random and trail-based deployment might provide some of the benefits of both approaches, by reducing individual heterogeneity whilst still providing sufficient data for modelling.

Herein, we examined sex ratios of five felid species and estimated density for four of them, as the capture rate for tiger was too low for reliable SCR modelling. Male and female felids are known to differ in habitat use due to sex-specific life-history strategies; here, we examined how these differences influence detection patterns under trail-based and random camera placements. Moreover, we expected that these differences in habitat use interact with the two placement strategies we used for the camera traps: trail-based (non-random) and random. Using simultaneous trail-based and random camera trap surveys overlaid on top of each other, we aimed to achieve three objectives: (1) to test whether male and female felids differ in their relative use of trails, with

females exhibiting lower detection rates on trails compared to males—patterns that may reflect, but do not directly test for, sex-specific differences in space use or behavioural avoidance; (2) to assess the extent to which trail-based and random camera placement approaches reduce apparent bias in density estimates, and (3) to use sub-sampling analyses of our data to provide survey design recommendations for cases where it is not possible to match the sampling intensities we used, which we refer to as ‘reduced-effort’ survey designs (either with fewer available camera traps or fewer sampling sites). These sub-sampling analyses included hybrid designs combining both random and trail-based cameras. The results of our study ultimately have broader implications for best practice in camera trap surveys of any species exhibiting individual heterogeneity, which probably encompasses a much wider set of species than are currently recognized.

2. Materials and methods

2.1. Study area and sampling design

We carried out fieldwork in Jigme Singye Wangchuck National Park and Royal Manas National Park, in south-central Bhutan. Our study area was defined as a 400 km² survey grid straddling the two national parks (Fig. 1). This area was divided into 100 contiguous spatial grid cells each of 2 × 2 km in ArcMap v.10.8.2 (ESRI, 2022). We adopted a 2 × 2 km grid size as a balance between ensuring sufficient spatial recaptures (necessary for SCR) and covering a sufficiently large area to encompass the home-ranges of sufficient individual cats (Tobler and Powell, 2013). We opted for two modes of camera deployment: trail deployment, and a superimposed random deployment (Fig. 1).

The trail-based deployment (which we refer to hereafter as the ‘100 T’ survey) comprised 100 stations with paired cameras (to capture both flanks of each individual), at a density of one station per 4 km² grid cell (Fig. 1). Each of these stations was located along a human or animal

trail, riverbed, or ridgeline, in line with the current convention for trail-based surveys.

The random deployment (which we refer to hereafter as the ‘100R’ survey) comprised 100 single-camera stations at the same density as trail-based locations but deployed as close as possible to the exact mid-point of each 4 km² grid cell (allowing a maximum of ±50 m in cases where it was necessary to avoid major obstructions such as rocks or trees or cliffs, an error covering no more than 0.06% of the 2 × 2 km grid cell). A very small proportion (4% camera stations) of random cameras were situated on animal trails and ridgelines, reflecting the rarity of these habitat features in the study area. The spatial regularity of the random design ensured that cameras sampled habitat randomly in the absence of systematic habitat blocks at the 4-km² scale.

In total, we deployed 300 camera traps across the 200 camera stations (‘100 T’ and ‘100R’ combined), which took approximately 9 weeks of fieldwork. Cameras were left in place for approximately four months, with some of the more accessible cameras checked halfway through.

The study focused on Bhutan's five felid species: tiger (*P. tigris*), leopard (*P. pardus*), clouded leopard (*Neofelis nebulosa*), marbled cat (*Pardofelis marmorata*), and leopard cat (*Prionailurus bengalensis*). Sex ratios were examined for five species, and density was estimated for four, as capture rates for tiger were insufficient for spatial capture–recapture (SCR) modelling.

2.2. Data preparation

We used image management software Digikam v.7.6 to annotate the camera trap images with species information. We then used the R package ‘camtrapR’ (Niedballa et al., 2016) to sort images by species. For individual identification, we first developed a reference catalogue from the trail-based stations using individuals for which both flanks were available, or where individuals could be reliably matched using other distinctive features such as tail markings, head patterns, or other

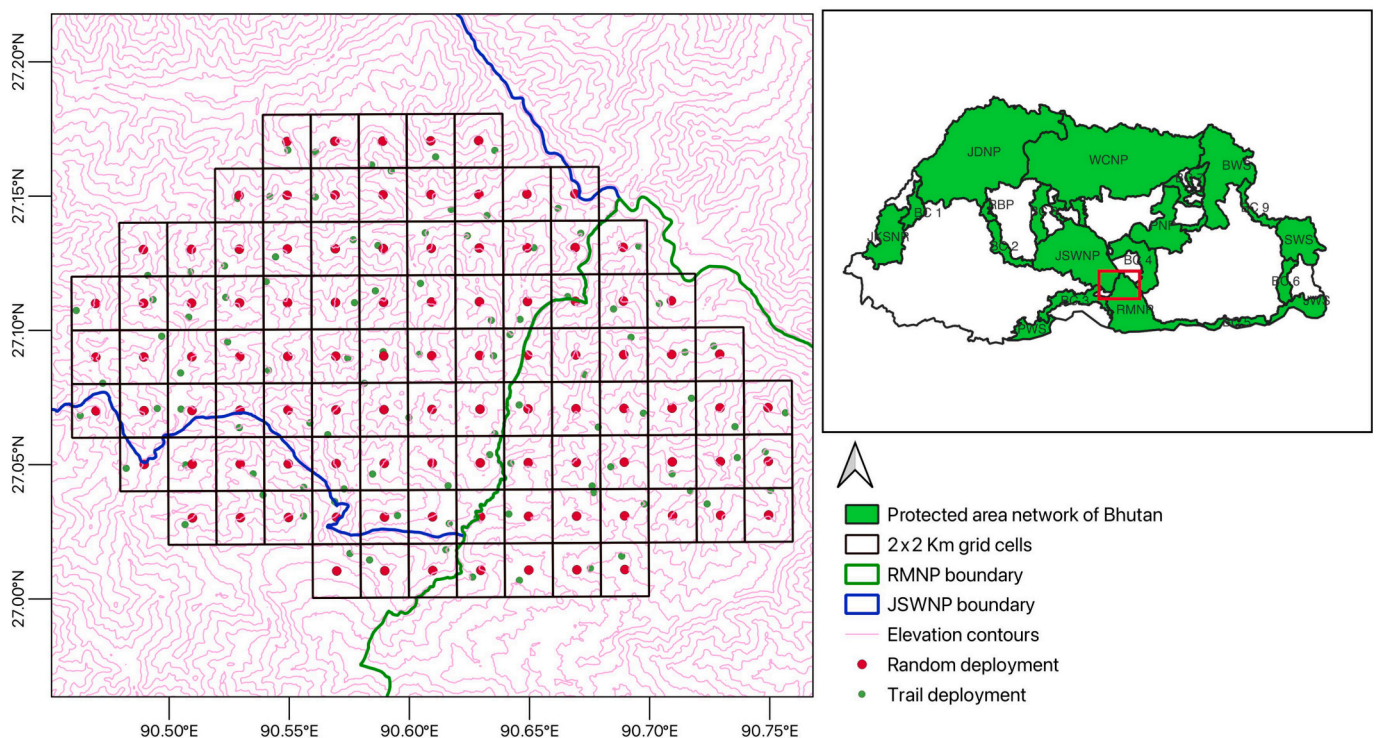


Fig. 1. Study area with overlay of 2 × 2 km grid cells with trail deployment (green dot) and random deployment (red dot). The study area falls in Jigme Singye Wangchuck National Park (JSWNP) and Royal Manas National Park (RMNP) and their buffers. The number on the contours shows the elevation in meters above sea level. The map on the right shows the protected area network of Bhutan. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

consistent pelage characteristics. All single-flank captures (both left and right) were then compared against this reference set. Where a high-confidence match could be made based on pelage features, the same individual ID was assigned. For individuals seen only partially that did not match the reference catalogue, new IDs were assigned only when individuals could be confidently distinguished based on multiple consistent features, including flank patterns, limb markings, facial features, and, where possible, repeated captures. Where uncertainty remained regarding whether left-only and right-only captures represented the same individual, those records were excluded and retained as “unknown” rather than being treated as separate individuals.

We then compared captures from the random cameras against this reference set using the same identification criteria. Individuals with a high degree of certainty in match quality were retained in the random camera dataset. The random cameras also yielded clearly new individuals, supported by images from multiple angles, that were not detected at the trail-based stations and were therefore added to the reference catalogue. In addition, some random camera captures allowed us to retrospectively identify individuals that had initially been classified as “unknown” in the trail-based dataset. We provide example images demonstrating the distinguishing marks we used for individual identification (Appendix S9), as well as a flow diagram of the approach we used (Appendix S8).

Despite our conservative approach to individual assignment, we acknowledge that some residual uncertainty may remain, particularly for individuals represented only by partial or single-flank images from random deployments. In rare cases, individuals photographed from different flanks may have been assigned separate identities if insufficient matching features were available, potentially resulting in false splitting of identities (‘ghost individuals’; Augustine et al., 2018; Kodi et al., 2024). Such errors could positively bias abundance and density estimates. However, we attempted to minimise this risk by excluding uncertain matches and requiring multiple consistent pelage features before assigning new identities.

We determined sex of each individual based on the presence or absence of external testicles and assigned each an individual identifier (Petersen et al., 2020). For individuals without a clear photograph of the inguinal region we classified them as of unknown sex. Only adult individuals were included in the sex-ratio and density analyses, as age could be reliably determined only for adults based on body size and pelage development. Juveniles and subadults were excluded from the final detection histories. To minimise potential bias arising from mortality, emigration, or immigration during the survey, we restricted the SCR analysis period to 60 consecutive days. The first day of the survey was defined for the entire trap array as 1 February 2021, thus accounting for the asynchronous camera trap deployment dates across stations (as specified in the R code of “camtrapR” package using `occasionLength = 60` and `day1 = “2021-02-01”`; Efford et al., 2009; Karanth and Sunquist, 1995; Piña-Covarrubias et al., 2023; Zimmermann et al., 2016). We acknowledge that, given our analysis window of 60 days, some degree of turnover cannot be entirely ruled out, and the assumption of complete population closure may therefore have been partially violated. This may have imparted a negative bias into capture probabilities and therefore positive bias into our density estimates.

2.3. Density estimation

We used the spatial detection histories prepared using camtrapR for model fitting, based on maximum-likelihood estimation in the ‘secr’ R package (Efford, 2023). For the buffer distance around the cameras (an input parameter for SCR), beyond which the detection probability of species approach zero (Efford, 2023; Piña-Covarrubias et al., 2023), we followed the recommendation of using around four times the initial estimate of σ (Efford, 2023). For clouded leopards, this resulted in a buffer distance of 25 km around the outermost trail and random camera stations. We used this distance for the other species, on the expectation

that they would have approximately equivalent, or smaller, home ranges. The total state-space, encompassing the detector grid (400 km²) and the 25 km buffer, covered approximately 2100 km², within which densities were estimated. The mesh spacing comprised equidistant points corresponding to the potential centre points of home ranges (Efford, 2023; Gopalaswamy et al., 2012). This corresponds to a grid of 250 × 250 m cells covering all suitable habitat within the 25 km buffer. The mesh spacing was selected to balance spatial resolution and computational efficiency and was kept constant across all species (Efford, 2023; Gopalaswamy et al., 2012; Penjor et al., 2018). In all SCR models, density was modelled as constant across the study area, as no spatial covariates (e.g., habitat type or elevation) were included to estimate a density surface. The modelling therefore focused on accounting for individual heterogeneity in detection and space use through the inclusion of sex and mixture components in detection parameters, rather than on spatial variation in underlying population density.

We applied the half-normal detection function for all the models, which comprised two detection parameters: g_0 , the baseline detection probability at the home range centre, and σ , the decline of encounter rate as a function of distance from a home range centre (Royle, 2011; Tobler and Powell, 2013). We constructed models (detailed below) according to our three outlined objectives.

2.4. Sex-dependent habitat-use

We tested our hypothesis that females exhibited lower detection rates on trails than males. First, we used the observed data on the number of individuals seen in each of the two camera deployment types, uncorrected for imperfect detection, as an unconditional measure of occurrence at the two camera deployment types. This unconditional measure will be a function both of capture probability and density at the cameras. In particular, we note that the trail-based design used paired cameras, so we might expect capture probabilities, and therefore the number of individuals to be slightly higher than if single cameras had been used. Using the observed number of individuals, we calculated the observed sex ratio (overall, and for each of the identifiable cat species) for the two deployment types. We investigated the sex ratio of individuals that were uniquely seen in either the trail-based or random survey. These individuals would have been missed by each of the surveys if conducted in isolation. Again, a male-biased sex ratio in the trail-only survey compared to the random-only survey would suggest that males were over-sampled by the trail survey compared to the random survey and females under-sampled.

We also tested our hypothesis through explicit modelling. Given the caveats associated with the observed data, the modelling approach was our primary focus when making inferences. Specifically, we used the full dataset (which we refer to hereafter as ‘100 T + 100R’) to fit a model with an interaction between a site covariate for deployment type (trail or random) and an individual covariate for sex (male, female, or unknown). This interaction was specified as acting on the g_0 parameter. Model selection among candidate formulations (sex-specific vs. pooled models) was evaluated using likelihood ratio tests (LRTs), comparing nested models that differed in the inclusion of sex as a covariate on g_0 or σ (Burnham and Anderson, 2002). The LRT statistic ($-2 \times \Delta \log L$) was compared to a χ^2 distribution with degrees of freedom equal to the difference in the number of parameters. As summarized in Table 1, we report the LRT results alongside AICc values to indicate whether inclusion of sex significantly improved model fit. If the interaction model was favoured, we interpreted this as support for sex-dependent habitat use. If, the g_0 estimates for females were lower on trails vs random locations, and males had the opposite pattern, we interpreted this as support for our hypothesis. Since we had a strong a priori basis for expecting differences in movement scale between the sexes, and because our focus in this study was on sex-dependent detection probability rather than movement scale, we included the individual covariate for sex on the σ parameter for all models. All models were fit using a hybrid mixture

Table 1

P-values from Likelihood Ratio Test (LRT) for density estimates of felids from four models: deployment (random, trail), sex (male, female), interaction (deployment * sex) and null applied to the full 100 T + 100R deployment. For example, $p = 0.017$ for the clouded leopard LRT comparing the interaction model versus (Vs) the deployment model, indicating that the interaction model fits better than the deployment model. Grey areas encompass models that are not comparable with each other.

Vs	Deployment	Sex	Null
Clouded leopard			
Deployment			0.004
Sex			0.006
Interaction	0.017	0.012	0.001
Null			
Leopard			
Deployment			0.34
Sex			0.30
Interaction	0.03	0.31	0.33
Null			
Leopard cat			
Deployment			0.00
Sex			0.00
Interaction	0.00	0.00	0.00
Null			
Marbled cat			
Deployment			0.00
Sex			0.03
Interaction	0.02	0.00	0.00
Null			
Tiger			
Deployment			0.00
Sex			0.47
Interaction	0.73	0.14	0.22
Null			

model, in the secr package (Efford, 2023). These models extend standard mixture models by allowing partial class membership information: individuals of known sex are assigned fixed class membership (male or female), whereas those of unknown sex have their membership estimated probabilistically from the data (Efford, 2023). The model therefore jointly estimates detection parameters (g_0 , σ) for each class and the proportion of individuals belonging to each class (the mixing parameter π), using maximum likelihood (Efford, 2023).

2.5. Quantifying apparent bias in density estimates

We evaluated density estimates from the trail-based survey (100 T), with the expectation that they might be negatively biased because some trail-avoiding individuals were never exposed to sampling and therefore remained undetected, in addition to the effects of individual heterogeneity. For this, we compared the 100 T density estimates against a reference estimate obtained from the combined 100 T + 100R dataset modelled with an interaction between deployment type and sex. Although we contend that estimates from the 100 T + 100R might also be biased due to individual heterogeneity in detection probability, the intensity of sampling and diversity of deployment sites used across the two surveys meant that this dataset provided the best chance of detecting all individuals in the survey area, which otherwise might be missed by the other surveys in isolation.

We made two further comparisons to tease apart the separate effects of undetected individuals and individual heterogeneity. For the first comparison, which aimed to assess whether undetected individuals contributed to bias, we compared density estimates from a null model for the 100 T dataset and the combined 100 T + 100R dataset. For the

second comparison, which aimed to isolate the effect of individual heterogeneity, we compared density estimates from the 100 T dataset using the null model and a model with sex-dependent detection parameters.

We additionally acknowledge the possibility of “ghost individuals” as defined by Kodi et al. (2024) whereby identification errors may result in the same individual being assigned multiple identities. Although we attempted to minimise such errors through conservative matching criteria and exclusion of uncertain records, some residual false splitting cannot be completely excluded, particularly for single-flank captures from random deployments.

2.6. Assessing analytical- and randomised approaches to dealing with individual heterogeneity

We next assessed whether analytical and randomised placement approaches to dealing with heterogeneity in individual detection might reduce bias in density estimates by comparing estimates under each approach with our reference estimate from the 100 T + 100R dataset (with an interaction between deployment type and sex specified on g_0 , as detailed in Section 2.4). The analytical approach involved fitting a sex-dependent model to the 100 T dataset, whilst the randomised placement approach involved adding random cameras into the survey design, i.e. 100 T + 100R with a null model fit to the data. We tested this for four cat species (clouded leopard, common leopard, leopard cat and marbled cat) excluding tiger due to insufficient captures.

2.7. Exploring ‘reduced-effort’ survey designs through sub-sampling

We estimated density from two alternative scenarios representing typical constraints on camera trap deployments: (1) limited availability of equipment, specifically camera traps, or (2) limited availability of personnel, time and/or funding resources. The reduced-effort datasets were created by randomly sub-sampling stations without replacement (Fig. 2).

To evaluate how reduced sampling effort might influence density estimates, we created a series of sub-sampled datasets from the full 200-station design (100 trail + 100 random stations). Two scenarios were designed to represent realistic field constraints (Fig. 2). Scenario 1 (equipment constraint) assumed a total of 100 available cameras. Cameras were allocated in alternative combinations of trail and random stations representing 50%, 67%, 75%, 88% and 100% of the total 200 stations. These combinations ranged from paired cameras at 50 trail stations to single cameras at 100 random stations (Fig. 2).

Scenario 2 (labour/time constraint) assumed a fixed effort of 100 grid cells but varied the proportion of trail- and random-based deployments among them. Configurations included 100% trail, 75% trail + 25% random, 50% trail + 50% random, 25% trail + 75% random, and the full 100 T + 100 R deployment (300 cameras total; Fig. 2; Appendix Table S1).

For each of the four species (i.e., excluding tigers, due to the limited sample size) and for each of 10 simulated surveys (five combinations considering equipment constraints and five combinations considering resource constraints), we constructed a total of 14 models and compared the resulting density estimates. Eleven of the models were the standard models provided in the secr package (Efford, 2023; Zimmermann et al., 2016), and the remaining three models included sex as an individual covariate (Appendix S1). For each species and each simulated survey, the best-supported models for density estimation were identified using Akaike's Information Criterion (AIC; Burnham and Anderson, 2002; Zimmermann et al., 2016). Note that we have assumed that all the individuals we were able to identify across our combined trail-based and random camera data would remain identifiable in these smaller sub-samples. This may mean our results are optimistic with respect to capture probabilities, unless reference image sets (e.g., from previous monitoring of the survey areas) are available.

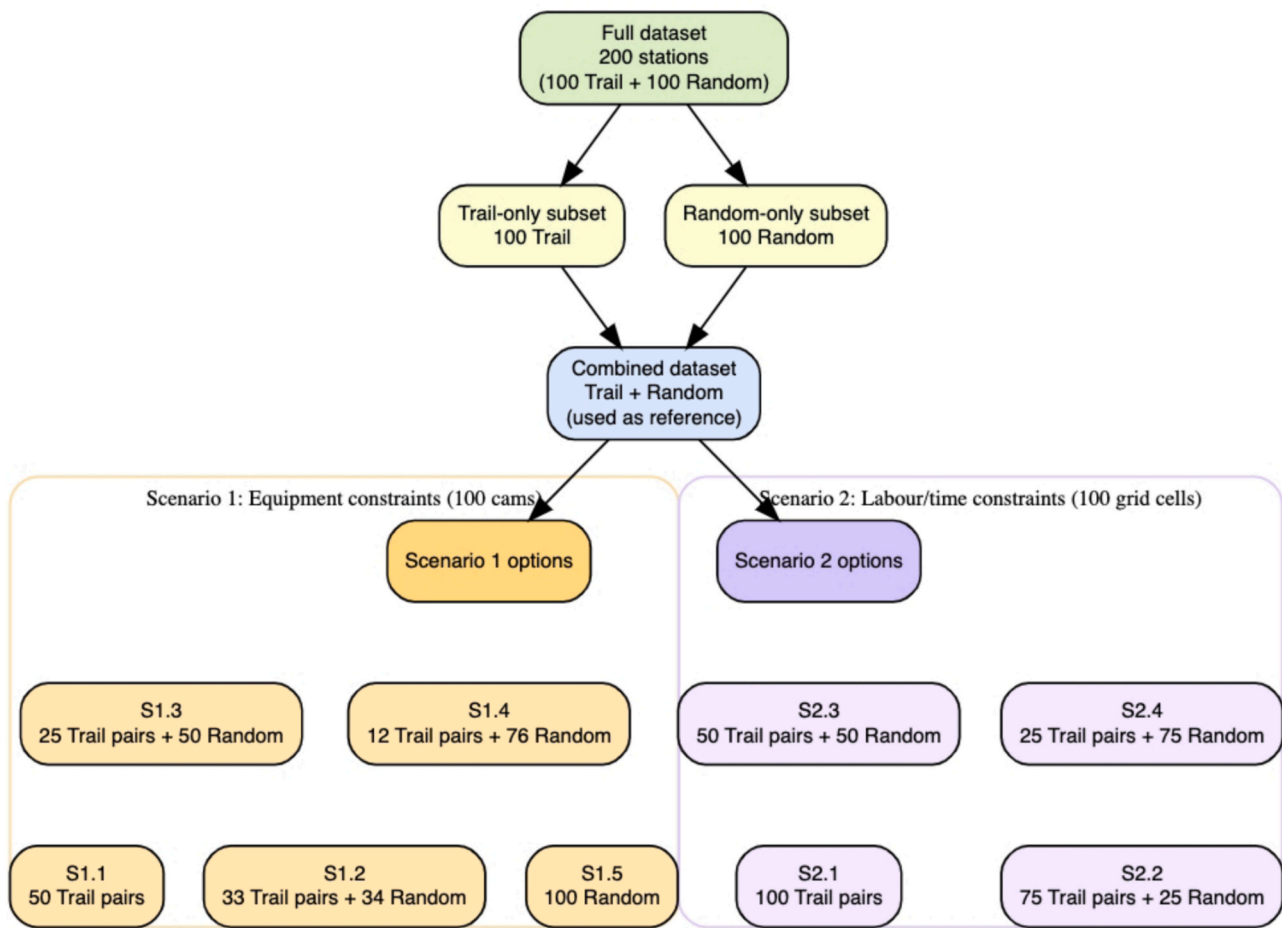


Fig. 2. Camera-trap dataset sub-setting and resulting deployment scenarios used for comparative analyses.

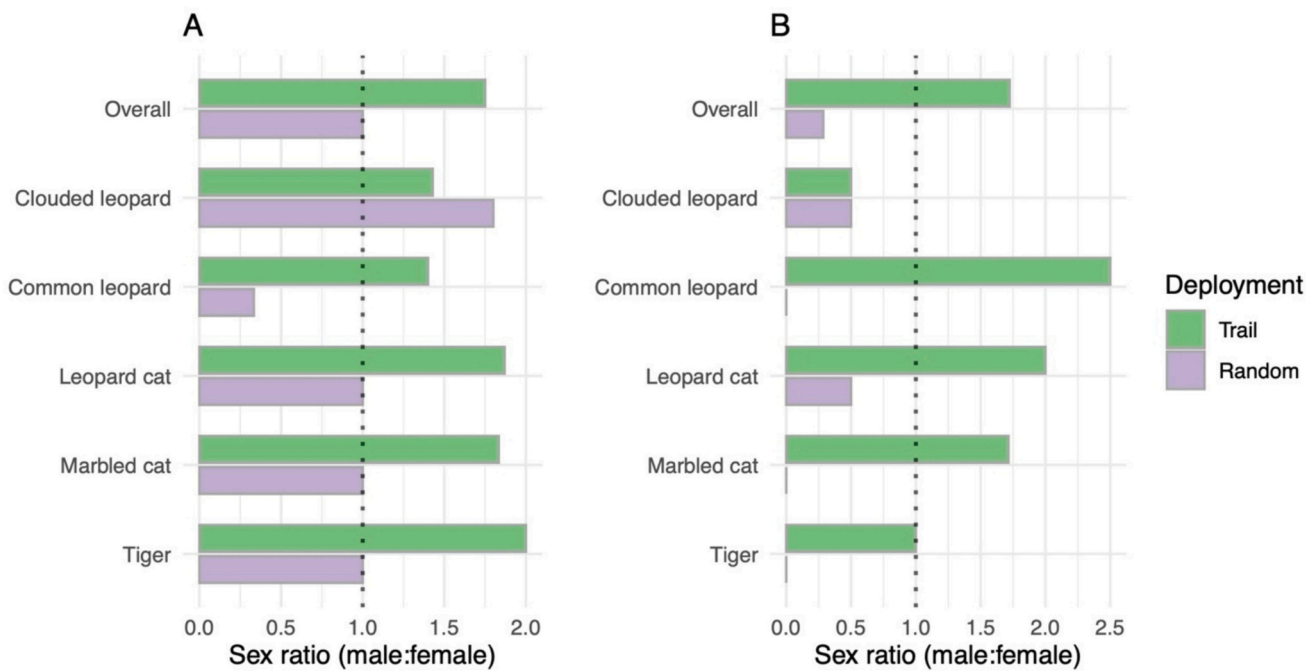


Fig. 3. Male-to-female ratios observed in trail-based and random camera trap surveys, with panels showing overall male-to-female ratios (A) and sex ratios for individuals unique to a given deployment type (B).

3. Results

3.1. Observed sex ratios in the trail-based and random surveys

From 13th October 2020 to 5th May 2021, camera stations were operational for a total of 23,592 trap-nights (counting paired camera traps as a single unit). As well as our five focal species, we also detected two other felids that are not individually identifiable: *Catopuma temminckii* and *Felis chaus*. Across the five focal species, a total of 2496 independent photographic captures were obtained, of which 2332 (93.4%) could be confidently identified to individual level. The remaining 164 (6.6%) photographs were excluded from further analysis because individuals could not be uniquely identified due to partial body images, motion blur, and poor lighting. The proportion of unidentified captures varied among species, ranging from 0% for tigers, 1.1% for clouded leopards, 3.6% for leopard cats, 9.1% for marbled cats, and 19.8% for leopards.

Across the whole study and the five felid species (Table 1), we observed 90 males and 69 females, giving a male-biased sex ratio of 1.3. This male bias was substantially higher in the trail-based survey (100 T dataset), with a sex ratio of 1.75 (84 males and 48 females). The random survey (100R dataset) yielded an even sex ratio of 1.00 (40 males and 40 females). All species except clouded leopard exhibited a more strongly male-biased sex ratio in the trail-based survey versus the random survey (Fig. 3, Appendix S3).

A total of 50 males were exclusively observed in the trail-based survey (56% of the observed male population); only six males were observed exclusively in the random survey. This discrepancy was most pronounced for the leopard and leopard cat, in which 71% and 63% of males, respectively, were observed only on trails and apparently avoided the random deployments altogether (Euler plots depicting the number of unique males and females in each survey are provided in Appendix S2). By contrast, a large number of females were seen only in the random survey (21 females; 30% of the observed female population; Appendix S4). Nevertheless, more females were unique to the trail-based survey compared to the random survey (29 versus 21 females, respectively), owing to the overall higher capture probabilities in the trail-based

survey compared to the random survey. The sex ratios of unique individuals for the trail-based and random surveys were 1.72 and 0.29, respectively, further emphasizing the male-biased ratio for the trail-based survey and the female-biased ratio for the random survey.

3.2. Models of sex-dependent habitat-use

Sex-dependent habitat use was supported in three of the five felid species. Using the full 100 T + 100R dataset, likelihood ratio tests indicated that the interaction between deployment type and sex was a significant component of the density models for the clouded leopard *Neofelis nebulosa*, leopard cat *Prionailurus bengalensis*, and marbled cat *Pardofelis marmorata*, with only limited support for leopards *P. pardus* and tigers *P. tigris* (Table 1), although for tigers there was likely limited statistical power to detect a true effect.

Detection probabilities at home-range centre, g_0 , for the random deployment were 41% lower overall than in the trail-based survey (mean g_0 estimates were 0.0018 and 0.0030, respectively). For males, g_0 estimates were 56% lower in the random vs trail-based survey (g_0 of 0.0018 and 0.0040, respectively), whilst for females the difference was not as stark, at 12% lower (g_0 of 0.0018 and 0.0021, respectively). For all five species, detectability was higher for males on trails than at random locations (with the difference being statistically clear in leopard cats and marbled cats, Fig. 4). For females, two species showed a marginally higher g_0 at random locations compared to trail locations (leopard cats and marbled cats), whilst the other two species showed the opposite pattern, but none of these differences were statistically discernible (Fig. 4).

3.3. Density estimates for the trail-based and combined surveys

For three of the four felids (excluding tigers because of too few captures), the trail-based survey underestimated density compared to the reference from the combined dataset modelled with the interaction effect (Fig. 5 and Appendix 5). The percentage differences were largest for clouded leopards (−28%) and marbled cats (−10%). For leopard cats, the trail-based survey produced an estimate that was higher

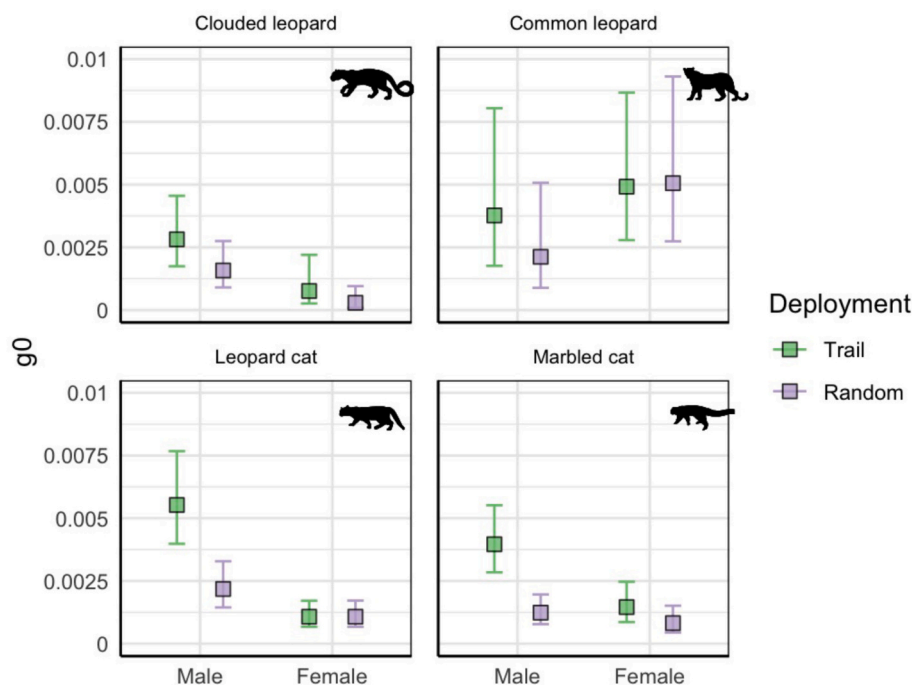


Fig. 4. Sex-dependent estimates of g_0 – detection probability at the home-range centre – for trail-based and random camera trap surveys across four felid species. Error bars show 95% confidence intervals.

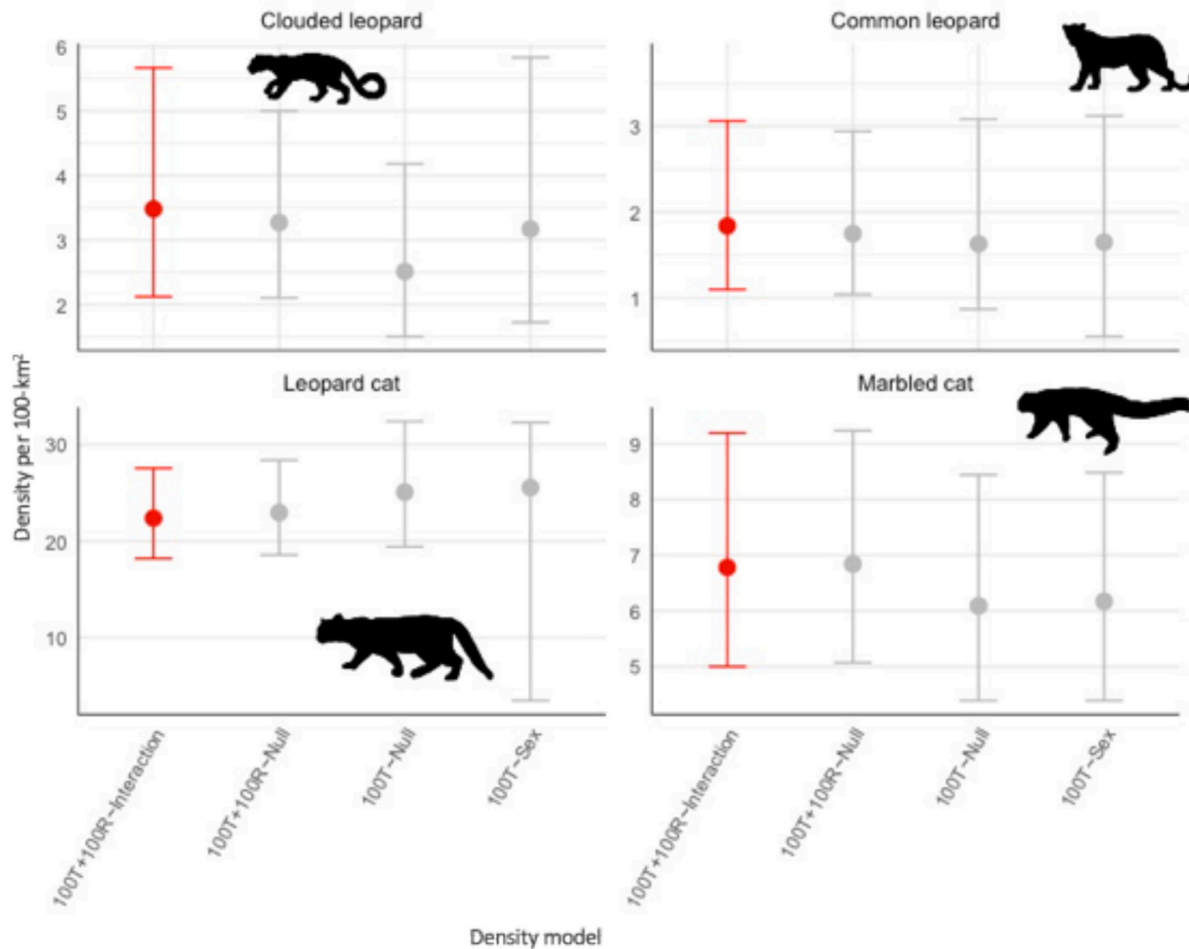


Fig. 5. Density estimates and 95% confidence intervals are shown for four models: 100 T + 100 R ~ Interaction (combined dataset with an interaction between deployment type and sex), 100 T + 100 R ~ Null (combined dataset with no covariates), 100 T ~ Null (trail-only dataset with no covariates), and 100 T ~ Sex (trail-only dataset with sex as a covariate), with error bars representing 95% confidence intervals.

(+12%) than for the combined dataset, despite the fact that 15 individuals (including 10 females) were missed by the trail-based survey.

In comparison, the density estimates from the 100 T and 100 T + 100 R datasets both modelled using a null model indicated that undetected individuals likely contributed to negative bias in the trail-based survey, with lower density estimates for the 100 T dataset than for the combined dataset for each of the felid species, except the leopard cat (Appendix 5). By comparing the density estimates for the 100 T dataset when modelled using the null model versus sex-dependent parameters, we also uncovered evidence of negative bias due to individual heterogeneity. This was apparent in all four felid species, although the density differences in the case of leopard cats were marginal (Appendix 5).

3.4. Analytical and randomised placement approaches to dealing with heterogeneity

We found that both analytical and randomised placement approaches were effective in reducing biases caused by individual heterogeneity in detection probability and movement. For three of the four felid species (clouded leopards, leopards and marbled cats), incorporating sex-dependent detection into the modelling of exclusively trail-based data appeared to correct biases in density estimates, making estimates indistinguishable from the reference estimates calculated using the full dataset (Fig. 5 and Appendix 5). In contrast, for leopard cats, the trail-based density estimate remained higher than the reference estimate, although there was considerable overlap in the confidence

intervals. The randomised placement approach was also effective for three of the four species (leopards, leopard cats and marbled cats), even when a simple null model was used (Fig. 5 and Appendix 5). For clouded leopards, the density estimate was lower than the reference estimate calculated using the full dataset, suggesting that an analytical approach was necessary for this species (although confidence intervals did overlap).

Compared with estimates derived from trail-only or random-only data, the combined trail + random design yielded density estimates that were less biased and also showed narrower confidence intervals, indicating improved precision. Thus, the principal gain was a reduction in apparent bias, accompanied by a moderate improvement in estimator precision.

3.5. 'Reduced-effort' survey designs

3.5.1. Constraints on availability of equipment

Overall, we considered density estimates from deployment of 100 T + 100 R to be the most accurate estimate for comparison between different combinations of trail and random camera stations. For clouded leopards, 100 R deployment yielded the highest density estimates in models without sex as a covariate, whereas the highest sex-covariate model estimates came from a combination of 12 trail and 76 random stations (Appendix S5). The 100 T deployment yielded slightly lower density estimates for no-sex models and significantly lower estimates for sex-based models. For leopards, the highest density estimates in no-sex

models were derived from 12 trail plus 76 random stations, while sex-based models peaked with 33 trail plus 34 random stations (Appendix S5).

For leopard cats, models with and without a sex covariate produced their highest density estimates with 33 trail and 34 random deployments, closely aligning with 100 T + 100R deployment results (Appendix S5). Similarly, marbled cats exhibited higher density estimates with 100R deployment, but these estimates had large standard errors and wide confidence intervals, reducing precision compared to 100 T + 100R or trail-only deployments (Appendix S5). Sex-based detections in marbled cats showed contrasting patterns, with random deployments favouring females and trail deployments favouring males. In general, increasing randomization improved the accuracy of density estimates across species, despite variability in sex-based detection ratios.

3.5.2. Constraints on availability of resources

With camera trap station availability limited to 100 grid cells, the model without a sex covariate produced the highest estimated density for clouded leopards (Appendix 6). When sex-based detection covariates were included, the 75 trail plus 25 random deployments yielded the highest density estimates (Appendix S6). Density estimates were slightly higher for most of the trail- and random-based station combinations with the 100-cell limit than for the estimates from the 100-trap limit, unsurprisingly since more stations allow more individual detections (Appendix S6).

For leopards, density estimates slightly increased with more randomization of camera stations. The highest density estimates for both no-sex and sex-based models were obtained from 50% random and 50% trail deployments. Even with a 100-camera limit, deploying more cameras at random stations yielded higher density estimates than deploying camera traps over a larger area of 100 grid cells. Distributing 100 cameras, with 12 trail-based and 76 random deployments yielded 2-fold higher density estimates compared to distributing 175 cameras across 75 trail 25 random deployments (Appendix S6).

Leopard cat density estimates were highest from a full trail deployment, regardless of the inclusion of sex parameters (Appendix S6), and were higher than for all estimates with the 100-camera limit (Appendix S6) and close to the estimate for full deployment (100 T + 100R). For marbled cats the highest density estimates were obtained from 75 trail-based and 25 random deployments, for all models regardless of the inclusion of a sex covariate. Estimates from the 100-cell limit tended to have smaller SEs than those from the 100-camera limit, reflecting the larger sampling area (Appendix S6).

4. Discussion

We assessed the presence of detection heterogeneity arising from sex-based differences in the behaviour of wild felids and its influence on density estimates. Our analyses revealed widespread sex-based detection heterogeneity reflected as differences in density estimates when camera traps are placed on trails compared to random locations. These results have repercussions for felid monitoring, as well as terrestrial mammal monitoring. Owing to limited sample sizes in some studies, an assumption is made that all individuals have an equal probability of detection whether those detections are made using camera traps, DNA from hair (Rounsville Jr et al., 2022) or faeces (Chettri et al., 2019) or more recently, sound (Growcott et al., 2025; Foster and Harmsen, 2012). Heterogeneity in sex-based detection probability generally lowers the accuracy of abundance estimates (Harmsen et al., 2010; Piña-Covarrubias et al., 2023), and unless accounted for, will produce biased results. Our comparison of random to trail-based deployments allowed us to test the equal-capture assumption, and thereby to explore the causes of bias in abundance estimates. Because density was modelled as spatially constant, our estimates represent mean density across the sampled landscape. In heterogeneous habitats, this assumption may mask local density variation related to habitat quality or resource

distribution (Efford, 2023; Royle et al., 2013). Future studies incorporating spatial covariates or density-surface modelling could help resolve fine-scale variation and improve biological interpretation of density patterns.

4.1. Male-biased density estimates from trail deployment

Male-skewed detection in trail deployment was observed for all felid species except for clouded leopards and tigers (although for the latter we had a small sample size). Our results suggest that males were detected on trails more frequently than females, with the propensity for males to be captured on trails leaving few male individuals to be detected only at random stations. In contrast, more females were detected only at random stations in comparison to trail stations. Male preferences for trails and females for off-trail habitat have been observed in other cat species, for example jaguars (Harmsen et al., 2020; Montalvo et al., 2023). Males tend to minimise their energetic costs by using linear features such as natural trails, mountain ridges, and rivers, which may also increase the probability of encountering prey (Harmsen et al., 2010; Wearn et al., 2013). Females, on the other hand, tend to avoid such features, presumably to avoid potential conflicts with conspecifics and reduce the risk of infanticide (Balme et al., 2009; Harmsen et al., 2017; Srbek-Araujo, 2018; Singh et al., 2014). Although numerous studies have suggested that felids preferentially use trails (Smith et al., 2023; Oliveira et al., 2018; Di Bitetti et al., 2006; Penjor et al., 2019), surprisingly little empirical evidence exists demonstrating this preference, and there is less information about sex-based differences in the relative use of trails and other linear features. Based on the number of individual female felids captured at randomly placed off-trail camera traps, and the number of male felids captured at non-random trail-based camera traps, our study provides some of the first evidence that some felids exhibit sex-based differences in the use of trails.

Our comparison between trail and random deployments demonstrates clear differences in detection patterns between sexes; however, inferences about behavioural avoidance of, or attraction to, trails should be made cautiously. Our observed sex-ratio patterns are a function of both imperfect detection and habitat use, and the single cameras in the random design may have contributed to the lower number of individuals observed compared with the trail-based deployments, which used paired cameras. In addition, we identified individuals using both datasets in combination (starting with the trail-based data), and the single cameras in the random design may have had even lower capture probabilities had no reference set from trail-based data (or previous surveys in the area) been available.

4.2. Detection heterogeneity in different sexes of wild felids

The assumption of equal capture probabilities of all animals within the study area (Pollock, 1982) is the main source of bias in density estimates from trail-based deployments of camera traps (Otis et al., 1978; Pledger and Efford, 1998). Our analyses of concurrent trail-based and random deployments revealed the presence of an interaction between sex and deployment type. Failing to include the sex-specific detection parameters in a density model can produce a negative bias, thereby underestimating density. Unsurprisingly we observed higher density estimates from models of sex-specific detection combined with the inclusion of random (off-trail) camera traps, which effectively reduced detection heterogeneity by capturing more females. Others have previously reported that most studies on density estimates of big cats lack proper descriptions of study design and data quality, as they lack information on the number of individuals, sex ratios, and the number of recaptures from which to estimate heterogeneity in the capture histories (Efford, 2014; Foster and Harmsen, 2012). Solely placing camera traps on trails and other linear openings has similarly been found to bias density estimates in rodents (Kolowski and Forrester, 2017).

Our findings are consistent with earlier research demonstrating that

camera trap placement can substantially influence estimates of population parameters in large mammals. In tropical systems, Fonteyn et al. (2021) found little difference in mammal diversity between trail-based and systematically placed cameras in Gabon, whereas Greco et al. (2025) reported that trail-based designs may bias community-level metrics by oversampling species that preferentially use trails. These contrasting outcomes suggest that the influence of camera placement is parameter-dependent: while trail placement may underestimate densities for species showing sex-specific or behavioural avoidance, it may have weaker—or even opposite—effects on parameters such as occupancy, species richness, or activity patterns.

Based on our results, we advocate a mixed approach to camera trap study design wherein cameras are placed both on trails and randomly to sample the entire population and reduce the likelihood of biased results. We caution, however, against placing all camera traps randomly, as this would not only underestimate the number of males but would also reduce estimate precision owing to fewer individual captures. Similarly, it is possible that the standard approach of placing all camera traps along trails may be suitable if it results in sufficient individual captures. However, doing so is risky as it is impossible to predict when or if a dataset will be sufficiently large to allow estimation of sex-specific detection heterogeneity.

4.3. Need for adequate survey design to reduce undetected individuals and identification errors

Our results indicate that the design of most felid camera trap studies means that they are likely to have missed detections of certain individuals and ultimately produced biased density estimates. In our study, trail-based camera deployments missed approximately 17% of all identified individuals (predominantly females), whereas randomly placed cameras missed about 47%, mostly males. A typical gridded survey design deploys camera traps inside grid cells approximately equal in size to the area of the focal species' home range (O'Connell and Bailey, 2011). We observed an increase in density estimates for clouded leopards, common leopards, and marbled cats when combining capture histories from both trail-based and randomly placed camera traps. This increase was primarily due to the addition of previously undetected individuals and may also result from a reduction in capture heterogeneity.

Beyond detecting new individuals, the number and spatial distribution of recaptures are equally critical for accurate parameter estimation in spatial capture–recapture models. Reliable estimation of detection probability (g_0) and movement scale (σ) requires that individuals be detected at multiple camera locations. Sampling designs that promote spatial recaptures—for example, by combining trail-based and random placements—enhance precision and reduce bias in density estimates. Moreover, maximizing survey effort to obtain repeated detections helps minimise false identifications, which can introduce substantial bias into capture histories. Kodi et al. (2024) demonstrated that careful validation of individual IDs, or exclusion of single-capture individuals, can substantially reduce this error and improve model robustness. In capture–recapture literature, the term ‘ghost individuals’ refers specifically to false duplicate identities arising from identification errors, such as mismatched partial flanks or genotyping errors (Kodi et al., 2024). Although we attempted to minimise such errors through conservative matching criteria, residual false splitting cannot be completely excluded, particularly for partial images from single-camera deployments.

Our findings highlight that while high-density trail-based camera deployments may be the standard, the addition of randomly placed cameras was crucial in detecting these otherwise unnoticed individuals. Our results also indicate that differences in trail use were not only observed between sexes of the same species, but also between species. This mirrors results that show jaguars and pumas occupying relatively similar niches likewise have been found to have different preferences for

trails, with pumas having a higher propensity to use trails and jaguars a higher propensity for off-trail movement (Harmsen et al., 2010). Although pumas (*Puma concolor*) are generally considered subordinate, it is interesting to consider whether in ecosystems with multiple felids, subordinate species may avoid trails to avoid dominant competitors and potential predators. In particular, individuals such as non-breeding or transient (“floater”) cats—often more wary of dominant conspecifics—may preferentially use off-trail areas and thus be under-represented in trail-only surveys. Such behavioural segregation between territorial and non-territorial individuals is consistent with spatial ecology studies in carnivores showing sex- and status-based differences in movement and space use (Mengülluoglu et al., 2021). The addition of random cameras therefore serves to better sample such otherwise under-detected individuals, increasing the representativeness of detections across the population (Findlay et al., 2020).

4.4. Insights on the design of camera trap surveys

Although fully randomised survey designs are often desirable for reducing systematic sampling bias in ecological studies (Quinn and Keough, 2002), camera trapping is laborious and resource intensive, often requiring a compromise between practical and ideal design (Cusack et al., 2015; Montalvo et al., 2023). Furthermore, completely randomizing camera trap placement failed to detect a higher proportion of individuals—approximately 47%—compared to about 17% missed when cameras were restricted to trails. As a compromise between the ideal and the practical, and to ensure sufficient sample size whilst also mitigating potentially biased density estimates, we recommend coupling random and trail-based deployment of camera traps.

Models including sex-specific detection probability increased density estimates when camera traps were restricted to trails. For clouded leopard, common leopard, and leopard cat, estimates from both types of trail-only deployment (camera-limited 50 T + OR and cell-limited 100 T + OR) increased, as did the density estimate for marbled cat for the cell-limited 100 T + OR deployment. Our results parallel other studies where the inclusion of sex as a covariate improved density estimates for jaguars when compared to the density estimates derived from null models (Sollmann et al., 2011). The accuracy of the density estimates was further improved with the addition of random deployments, although using only random deployments yielded too few captures for precise density estimation. To improve both the accuracy and the precision of density estimates, a combined approach of trail and random cameras is needed. We expect that this will be especially important in large-scale surveys, such as regional- or national-scale felid surveys, where the objectives are to obtain an estimate that is as precise and accurate as possible; the precision of small-scale surveys is usually poor (with coefficients of variation often approaching 100%), so in practice controlling for individual heterogeneity is unlikely to make a meaningful difference to estimates. Importantly, our recommendation to incorporate random camera placements should not be interpreted as support for single-camera deployments. Wherever possible, both trail-based and random stations should employ paired cameras to maximise bilateral identification and minimise false splitting of identities. The use of single cameras in our random deployment represented a logistical compromise to maximise spatial coverage with limited equipment. It is also justifiable where cameras are deployed at sites where animals may pause and interact with the environment in front of the camera, such as at natural marking sites. Paired cameras, though, remain the preferred configuration for SCR studies of individually identifiable species because they reduce uncertainty associated with partial identities and improve the reliability of individual assignment.

In addition, we recommend that survey design is informed by simulations, based on approximate estimates of density (D), detection probability (g_0), and σ . Such simulations can be done using the R packages ‘secrdesign’ (Efford, 2023) and ‘oSCR’ (Sutherland et al., 2019). Simulating data beforehand provides a greater likelihood that

data needs are met.

4.5. Cost-effective measures for improving accuracy of camera trap studies

Extensive camera trapping fieldwork incurs high costs in terms of equipment, time, and labour (Cusack et al., 2015). Cost increases with the number of sampling units (Tarugara et al., 2019), and surveys involving estimation of animal abundance are mostly focused on trade-offs between the number of sampling units and the number of sampling occasions and rarely account for the influences on estimation accuracy caused by placement of the sampling units (Gálvez et al., 2016). For survey areas like the one described here, availability of 100 camera traps can provide accurate density estimates if the survey design takes into consideration detection heterogeneity and proper simulation prior to beginning fieldwork. In particular we recommend that camera trap surveys move beyond trail-only deployments and integrate strategically randomised or spatially representative placements as a cost-effective way to improve the robustness of density estimates.

CRedit authorship contribution statement

Lungten Dorji: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **C. Patrick Doncaster:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wai-Ming Wong:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Rinjan Shrestha:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Kuenley Tenzin:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Paul Glover-Kapfer:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition. **Lon Grassman:** Writing – review & editing, Supervision, Methodology, Data curation. **Oliver R. Wearn:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Lungten Dorji reports financial support was provided by World Wildlife Fund. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biocon.2026.111987>.

Data availability

Data will be made available on request.

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